

Name: _____

Date: _____

PYTHAGORAS, (APPLYING PYTHAGORAS IN 2D PROBLEMS)

LO: I can apply Pythagoras' theorem to solve problems in 2D contexts.

Pythagoras in context

Pythagoras' theorem states $a^2 + b^2 = c^2$ where c is the **hypotenuse** (longest side, opposite the right angle).

Strategy: Identify the right-angled triangle in the problem, label the sides, decide if finding the hypotenuse (add) or a shorter side (subtract), then solve.

Quick examples

●	Distance between (0,0) and (3,4) = (9+16) = 5	coordinate distance
●	Diagonal of rectangle 6 by 8 = (36+64) = 10	add squares
●	Ladder 5m, base 3m from wall: height = (25-9) = 4m	subtract
●	Diagonal of 6 by 8 rectangle = 6 + 8 = 14	cannot just add sides

Key idea

Identify the right angle, label the hypotenuse (opposite it), then use $a^2 + b^2 = c^2$.

$$c = (\text{a}^2 + \text{b}^2)$$

Hypotenuse = square root of (sum of squares)

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - coordinate distance

Find the distance between A(1, 2) and B(4, 6).

Horizontal distance: $4 - 1 = 3$

Vertical distance: $6 - 2 = 4$

$$\text{Distance} = (3^2 + 4^2) = (9 + 16) = 25 = 5$$

5 units

The distance formula is just Pythagoras applied to coordinates.

Example 2 - diagonal of a rectangle

Find the diagonal of a rectangle 5 cm by 12 cm.

The diagonal is the hypotenuse of a right-angled triangle

$$d^2 = 5^2 + 12^2 = 25 + 144 = 169$$

$$d = \sqrt{169} = 13 \text{ cm}$$

13 cm

The diagonal splits a rectangle into two right-angled triangles.

Example 3 - ladder problem

A 10 m ladder leans against a wall with its base 6 m away. How high does it reach?

Ladder = hypotenuse = 10, base = 6, height = ?

$$h^2 = 10^2 - 6^2 = 100 - 36 = 64$$

$$h = \sqrt{64} = 8 \text{ m}$$

8 m

The ladder is the hypotenuse; subtract to find the shorter side.

Example 4 - isosceles triangle height

An isosceles triangle has base 10 cm and equal sides 13 cm. Find the height.

Height bisects the base: half-base = 5 cm

$$h^2 = 13^2 - 5^2 = 169 - 25 = 144$$

$$h = \sqrt{144} = 12 \text{ cm}$$

12 cm

Drop a perpendicular from the apex to create two right-angled triangles.

YOUR TURN – PRACTICE (deliberate practice)

1. Coordinate distances

Find the distance between each pair of points.

- a) (0, 0) and (3, 4)
- b) (1, 2) and (5, 5)
- c) (0, 0) and (5, 12)
- d) (2, 1) and (6, 4)

2. Diagonals

Find the diagonal of each rectangle.

- a) 3 cm by 4 cm
- b) 8 cm by 15 cm
- c) 7 cm by 24 cm
- d) 9 cm by 12 cm

3. Ladder and ramp problems

Solve each real-life problem.

- a) Ladder 13 m, base 5 m from wall. Height?
- b) Ramp 5 m long, rises 3 m. Horizontal distance?
- c) Flagpole 12 m, wire anchor 5 m away. Wire length?
- d) Slide 15 m, height 9 m. Horizontal distance?

4. Triangles and shapes

Find the unknown measurement.

- a) Isosceles triangle: base 8, equal sides 5. Height?
- b) Equilateral triangle side 6. Height?
- c) Square diagonal 10 cm. Side length?
- d) Rhombus diagonals 6 and 8. Side length?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) The hypotenuse is always the longest side ☐

Correct answer: _____

Reason: _____

b) $a^2 + b^2 = c$ means add and square root ☐

Correct answer: _____

Reason: _____

c) To find a shorter side, subtract the squares then square root ☐

Correct answer: _____

Reason: _____

d) Pythagoras works for all triangles ☐

Correct answer: _____

Reason: _____

e) The distance formula uses Pythagoras' theorem ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A rectangular field is **120 m by 50 m**. A diagonal path is built across it. How much shorter is the diagonal path than walking along two edges? Give your answer to 1 decimal place.

Expression: _____

Simplified: _____

b) Three towns A, B, C form a right angle at B. AB = **15 km** and BC = **20 km**. A new road is built directly from A to C. A fourth town D is at the midpoint of AC. Find the distance BD to 1 decimal place.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

